Co-Design Rankings Do Not Survive Realization: An 18-Morphology, 4-Environment Study

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***Abstract*—Learning-based co-design returns a *ranking* of candidate bodies scored in an abstract simulator whose cheapness makes the search tractable. We ask how much of that ranking survives the step that makes a design buildable. Using 18 morphologies of a modular icosahedral legged robot — a stratified sample of our own evolutionary co-design space — we train each with a shared PPO recipe in a replica of the abstract space the search used and in a fabricable space calibrated against CAD and motor datasheets. The two rankings are uncorrelated ($\rho_s = +0.11$, $p = 0.66$, mean $|\Delta\text{rank}| = 5.9$ of 18): the abstract winner falls to 7th, and a design paralyzed in the abstract space (0.54 m, 17th) comes 2nd at 10.8 m. The reshuffle survives a 6×3 seed replication ($P < 0.01$ under resampling), reward re-weighting and mass loading within the abstract space ($\rho_s \geq 0.77$), and halving the training budget ($\rho_s \geq 0.90$); a servo-level split attributes it to the geometry–DoF–rest-pose–mass half of realization rather than the actuator model ($\rho_s = -0.30$ vs $+0.61$). Changing the *task environment* instead preserves the ranking (rubble $\rho_s = 0.90$, ice 0.72); only discrete footholds are as disruptive. We trace the reshuffle to checkable mechanisms — a shell-sliding exploit, penetrating or floating rest poses, a leg-count advantage that exists only on paper — and argue that fidelity belongs inside the co-design loop, not after it.**

I. Motivation

Generative proposal of morphologies and iterative improvement by RL, optimization, or evolution [1]–[6] share an output format: a ranked shortlist of bodies, each scored by a policy trained in a deliberately abstract simulator. Our prior work on this platform is an evolutionary loop over icosahedral leg placements (a speciated EA in MuJoCo MJX [7]: seven DoF-partition species, 70 individuals $\times$ 40 generations $\times$ 20M steps). We do not re-run that search; we sample its space and ask what happens to the ranking when the same 18 bodies become machines that can be printed and bolted to Dynamixel XM430-W350 servos.

II. Platform and Protocol

Robot. ICOS is a 20-face icosahedral body (160 mm circumdiameter) with identical 3-DoF leg modules bolted to chosen faces; a morphology is which faces carry legs. The realized quadruped weighs 2.20 kg, stands 134 mm and carries 12 servos.

Morphologies. 4 hand-designed baselines plus 14 unevolved generation-0 samples, two per species: a stratified sample of the search space rather than of its optimum.

Training. One recipe, identical but for the entropy coefficient (0.01 abstract, 0.003 fabricable): Genesis [8] GPU simulation for both (the abstract space re-implements the search's MJX models), 4096 parallel environments, PPO (rsl-rl [9]), 2000 iterations (197M steps, 0.4–1.4 GPU-h, Apple M5 Pro), 20 s episodes, 50 Hz control over 250 Hz physics, reward $r = 2.5\,\Delta d + 0.5\,\text{alive} - 0.01\|a\|^2$. Locomotion is blind (no exteroception, no commands). We report mean displacement over 4096 rollouts — never the best of automatic retries — plus an *honest* variant truncated at the first fall (Sec. III-C).

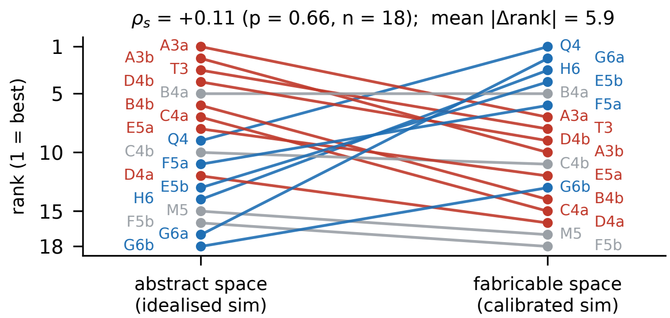

Fig. 1. Flat-ground ranking of the same 18 morphologies before and after realization. Blue rises, red falls; grey moves ≤ 2 ranks. Q4/T3/M5/H6: hand-designed quadruped, tripod, mixed, hexapod; A–G: the seven DoF-partition species from $(4, 4, 4)$ to $(2, 2, 2, 2, 2, 2)$, two gen-0 samples a/b each; the digit is the leg count.

Realization. Five things change. The freely partitioned 12-DoF budget (2–5 DoF per leg) becomes a fixed printable 3-DoF module (DoF $= 3\times$legs) — a manufacturing constraint: $(4, 4, 4)$ cannot be built from one printable module. Sphere proxies become a convex-hull body mesh with motor cylinders and skirt rings; the unreachable straight-leg rest pose becomes a solved, verified equilibrium ($\Delta z < 25$ mm, no penetration); the ideal position servo becomes torque PD ($k_p \in [25, 60]$) under a torque–speed envelope (4.1 N·m stall, 4.8 rad/s no-load) with 0–20 ms latency, sensor noise and a thermal penalty; and lumped masses become CAD part masses at their centroids (the quadruped's 2.2 kg is unchanged, but mass now scales with leg count: 1.8–3.1 kg and 9–18 servos for 3–6 legs). The last four are standard sim-to-real practice [10], [11], their parameters re-imported from CAD across seven hardware generations.

III. What Survives

A. The ranking does not survive realization

The two flat-ground rankings (Fig. 1) are statistically unrelated ($\rho_s = +0.11$, $p = 0.66$; mean rank shift 5.9 of 18, vs 6.0 expected if independent). The abstract champion (3 legs,

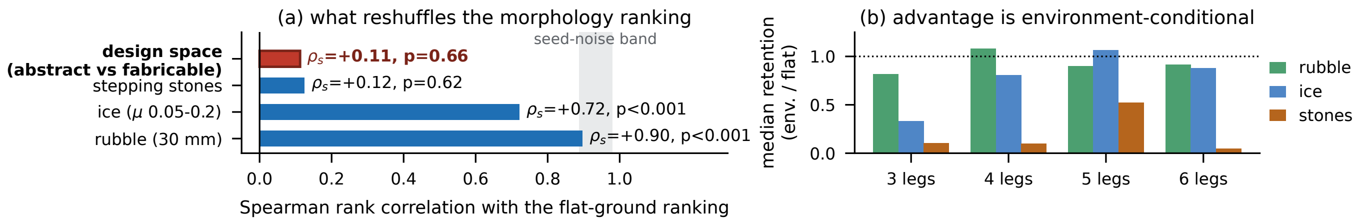

Fig. 2. (a) Rank agreement with the fabricable flat-ground ranking; shaded: the 5–95% band that seed noise alone produces on an unchanged ranking. (b) Median retention (environment / flat displacement) per leg-count group.

17.5 m) lands 7th at 5.53 m; the two worst abstract designs, paralyzed there (0.54 and 0.13 m), become 2nd (10.8 m) and 13th.

Seed noise cannot produce this. A 6-morphology × 3-seed replication on fabricable flat ground gives a median CV of 0.13 (range 0.03–0.63); reseeding moves the ranking only mildly ($\rho_s = 0.77$–0.89 between seed pairs). Under the null of an unchanged ranking perturbed only by measured seed noise, 20,000 resamples give $P(\rho_s \leq 0.11) < 10^{-4}$ at the median noise level and 0.006 at the noisiest.

Nor is it configuration or budget: raising the abstract reward's distance weight from 1.0 to 2.5 or loading +57% realistic part mass moves the ranking little ($\rho_s = 0.77$–0.97 between the four abstract variants), yet none of them agrees with the fabricable ranking ($\rho_s \leq 0.11$). In both spaces the half-budget ranking already matches the final one ($\rho_s \geq 0.90$; 0.84 at a quarter), and the disagreement is already present at half budget ($\rho_s = 0.21$).

The bundle also splits. Re-training the 18 fabricable bodies with the abstract space's ideal servos (no envelope, latency, noise or randomization) yields a ranking unrelated to the abstract one ($\rho_s = -0.30$, $p = 0.23$) yet close to the fully calibrated one ($\rho_s = +0.61$, $p = 0.007$): the geometry–DoF–rest-pose–mass half of realization destroys the ranking; the actuator layer perturbs it, unevenly (champion −13%; one design −55%, another +2.5×), hence 0.61 below the 5–95% seed band (0.89–0.98).

B. The ranking does survive the task environment

At fixed realization we retrained all 18 on quantized fractal rubble (30 mm), ice ($\mu \in [0.05, 0.2]$) and stepping stones (0.25 m stones, 50 mm gaps over a 0.5 m drop). Rank correlation with flat ground stays high (Fig. 2a): $\rho_s = 0.90$ for rubble, inside the seed-noise band, and 0.72 for ice, just below it (both $p < 0.001$). Only discrete footholds reshuffle like realization ($\rho_s = 0.12$, $p = 0.62$), because they change the problem: 15 of 18 morphologies learn to survive there (episode length >850 of 1001 training steps; standing still is valid) but only 4 of 18 exceed 1 m (median 0.49 m).

C. Mechanisms, not noise

Evaluation accounting is ruled out first: first-fall re-scoring leaves both endpoint rankings intact — the 18 abstract policies never fall (fall rate 0, $\rho_s = 0.94$ vs the mean ranking, 0.07 vs the fabricable one); fabricable first-fall agrees within 5% for 17 of 18; best-of-retries scoring had inflated only the earlier bad-rest-pose runs (5 of 18). Three checkable mechanisms remain. *(i) Exploits die.* The abstract 3rd-place tripod earned its 11.1 m by sliding its smooth spherical shell along the ground — legal under sphere-proxy collision, dead once the body is a convex-hull mesh with skirt rings. Realized, it walks — slower: 5.46 m, 8th. *(ii) The rest pose is a hidden design variable.* Of the 18 home keyframes, 6 penetrated the ground and 3 hovered. Re-solving them, morphology and reward untouched, moved flat-ground displacement 5.33 → 10.84 m and 5.60 → 10.09 m for two designs and 8.44 → 5.79 m for a third whose bronze medal had rested on a free fall; every result here uses the nine re-solved runs. *(iii) The abstract space rewards a trade hardware forbids.* Abstract displacement is predicted by leg count — fewer, longer legs win ($\rho_s = -0.89$), the tripod dominance our EA also found — and after realization the signal is gone ($\rho_s = +0.07$): the fixed module swaps the legs-versus-DoF trade for a legs-versus-mass one. Mirror symmetry, the EA's leading hypothesis, shows no significant relation in either space in this small sample ($|\rho_s| \leq 0.40$, $p \geq 0.10$); only the tripod baseline even has an icosahedral mirror plane.

D. Advantage is environment-conditional

No morphology dominates (Fig. 2b). Median retention on ice rises from 0.33 (3 legs) through 0.81 (4) to 1.06/0.88 (5/6), while rubble is nearly free (median 0.93) and stones ruinous (0.12).

IV. Implications and Limitations

A shortlist scored in an abstract simulator carries little information about the post-realization ordering — and the scoring step is the same whether candidates come from iterative search or zero-shot generation — yet does carry information across tasks: fidelity is part of the co-design objective. Treating realization as its own iterative loop kept this affordable: the 36 runs behind Fig. 1 cost an eighth of the search they audit.

Limitations. Runs outside the 6 × 3 replication are single-seed; only the aggregate contrasts, which survive the worst measured noise, are load-bearing. Within the geometry half of the split, DoF allocation, collision, rest pose and mass remain confounded (the split used the fabricable entropy). The abstract space is a Genesis replica of the MJX search environment, unchecked against it. No hardware data yet.

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